SPI1 设备文件配置方法

1、需要修改的内核设备树路径：

/arch/loongarch/boot/dts/loongson

2、在上面路径下需要修改或查看的文件有：

- loongon_2k0300.dtsi
- loongson_2k0300_pai_99_wifi.dts
- 2k0300-pinctrl.dtsi 三个文件

3、其中 loongon_2k0300.dtsi 文件中：

```
spi1: spi1@16018000 {
    status = "disabled";
    compatible = "loongson,ls-spi";
    #address-cells = <1>;
    #size-cells = <0>;
    reg = <0 0x16018000 0 0x1000>;
    spi-max-frequency = <50000000>;
    interrupt-parent = <&icu>;
    interrupts = <45>;
    num-cs = <1>;
    ls,bus_num = <1>;

    spidev@0{
        compatible = "rohm,dh2228fv";
        spi-max-frequency = <50000000>;
        reg = <0>;
    };
};
```

```
spi2: spi2@1610c000 {
    compatible = "loongson,ls-spi-v1";
    status ="disabled";
    #address-cells = <1>;
    #size-cells = <0>;
    reg = <0 0x1610c000 0 0x1000>;
    spi-max-frequency = <50000000>;
    interrupt-parent = <&icu>;
    interrupts = <6>;
    num-cs = <1>;
    ls,bus_num = <1>;

    spidev@0{
        compatible = "rohm,dh2228fv";
        spi-max-frequency = <50000000>;
        reg = <0>;
    };
};
```

该文件中，SPI1 的配置对照着 SPI2 复制就行，把图片中红框框选的部分修改成 SPI1 对应的值即可。需要修改 SPI1 对应的寄存器基地址，第修改 SPI1 对应的中断号，修改 compatible 的属性。

arch > loongarch > boot > dts > loongson > loongson_2k0300.dtsi

```
/dts-v1/;
#include <dt-bindings/interrupt-controller/irq.h>
#include <dt-bindings/gpio/gpio.h>
#include <dt-bindings/pinctrl/ls2k300-pinfunc.h>
/ {
    model = "loongson,generic";
    compatible = "loongson,loongson3";
    #address-cells = <2>;
    #size-cells = <2>;

    aliases {
        ethernet0 = &gmac0;
        ethernet1 = &gmac1;
        serial0 = &cpu_uart0;
        serial1 = &cpu_uart1;
        serial2 = &cpu_uart2;
        serial3 = &cpu_uart3;
        serial4 = &cpu_uart4;
        serial5 = &cpu_uart5;
        serial6 = &cpu_uart6;
        spi0 = &spi0;
        spi1 = &spi1;
        spi2 = &spi2;
        i2c0 = &i2c0;
        i2c1 = &i2c1;
        i2c2 = &i2c2;
        i2c3 = &i2c3;
        pwm_gtim = &pwm_gtim;
        pwm_atim = &pwm_atim;
    };
```

随后还在这个文件中，拉到最上方图片位置，添加上红框位置的程序即可。

4、其中 loongson_2k0300_pai_99_wifi.dts 文件中：

```
&spi1{
    status = "okay";
    pinctrl-names = "default";
    pinctrl-0 = <&spi1_4bit>;
};
```

```
&spi2{
    status = "disabled";
    pinctrl-names = "default";
    pinctrl-0 = <&spi2_pin>;
};
```

该文件中需要引出 SPI1 的节点，也可以对照 SPI2 写，需要将 status 修改为 okay，另一个需要修改 pinctrl-0，其中填入的参数可以在 2k0300-pinctrl.dtsi 中寻找。

5、其中 2k0300-pinctrl.dtsi 文件中：

```
    spi1_pin: spi1-pin{
        spi1_4bit: spi1-4bit{
            loongson,pinmux = <&gpa3 12 15>;
            loongson,pinmux-funcsel = <PINCTL_FUNCTIONMAIN>;
        };
        spi1_7bit_f0: spi1-7bit-f0{                         /* SPI1 have four cs */
            loongson,pinmux = <&gpa4 14 15>;
            loongson,pinmux-funcsel = <PINCTL_FUNCTION2>;
        };
        spi1_7bit_f1: spi1-7bit-f1{                         /* SPI1 have four cs */
                    loongson,pinmux = <&gpa5 0 0>;
                    loongson,pinmux-funcsel = <PINCTL_FUNCTION2>;
                };

    };
```

该文件不需要修改内容，查看图片位置，该位置存放 SPI1 引脚信息以及复用模式。

6、完成上面步骤，重新编译内核并部署就行，在/dev 目录，会生成对应 SPI 设备文件

```
[root@LS-GD test_app]# ls /dev/spidev1.0
/dev/spidev1.0
```